Breast cancer

Management by **receptor subtype** (HR+/HER2–, HER2+, TNBC — and now **HER2-low / ultralow**) crossed with **stage and setting** · AJCC 8th anatomic + prognostic staging

Compiled from the Stanford Oncology Handbook 2025, Breast Cancer (Schultz C, Khan S, Sidhu K, Yang M, Dalal N, Tsai J) · NCCN · primary trial reports verified against PubMed, September 2026.

WORKUP

- **Bilateral diagnostic mammogram + ultrasound**; MRI or contrast-enhanced mammogram if dense tissue, young, or extent unclear
- **Core biopsy of breast (and suspicious node) with clip placement** — clip the node before neoadjuvant therapy
- **ER/PR, HER2 (IHC, reflex ISH if 2+), grade, Ki-67** on every invasive cancer
- **Stage I–IIB: no routine staging imaging** — CBC/CMP; bone scan, CT only for symptoms, raised alk phos or abnormal LFTs
- **Clinical stage III: CT C/A/P + bone scan, or PET/CT**
- **Stage IV / recurrence: biopsy the first metastasis** (receptors drift, HER2-low status can change); brain MRI for CNS symptoms · **NGS on tissue or ctDNA** for PIK3CA/AKT1/PTEN/ESR1 · germline testing if not done
- Fertility counseling before chemotherapy in premenopausal patients

STAGING — AJCC 8TH

T1 / **T2** / **T3**	≤20 mm (T1mi ≤1, T1a ≤5, T1b ≤10) / >20–50 mm / >50 mm
T4	a chest wall · b skin edema/ulcer/satellite nodules · c both · **d inflammatory**
pN1 / **pN2** / **pN3**	1–3 / 4–9 / **≥10** axillary nodes (or infraclavicular, supraclavicular). pN1mi >0.2–2 mm; pN0(i+) ≤0.2 mm
cN2 / **cN3**	fixed/matted axillary or IMN alone / infraclavicular, IMN + axillary, or supraclavicular
Prognostic stage	**Grade, ER, PR, HER2 modify the anatomic stage** — triple-negative is upstaged, HR+/HER2+ often downstaged. **T1–2 N0 HR+/HER2– with Oncotype RS <11 → stage IA**. Pathologic prognostic stage does not apply after neoadjuvant therapy
After neoadjuvant	**yc** / **yp** prefix; pCR = **ypT0/is ypN0** — drives adjuvant escalation

DCIS: lumpectomy + RT, mastectomy, or **lumpectomy alone if small and age >70**; margin ≥2 mm; ER+ → tamoxifen ×5 yr (NSABP B-24) or low-dose 5 mg ×3 yr. **LCIS:** florid/pleomorphic → excise to negative margins; classic does not need them.

LOCAL THERAPY

Question	Evidence
1–2 positive SLN, BCS + RT	**Skip ALND** (Z0011: 10-yr OS 86.3 vs 83.6%). **AMAROS:** axillary RT = ALND for control (10-yr axillary recurrence 0.93 vs 1.82%), **half the lymphedema**
cN0, negative axillary US	**SLNB can be omitted**: SOUND (≤2 cm, 5-yr DDFS 98.0 vs 97.7%) · INSEMA (≤5 cm, BCS; 5-yr iDFS 91.9 vs 91.7%) — axillary recurrence slightly higher
RT after BCS, age ≥65, low risk	**PRIME II** (HR+, N0, ≤3 cm, on ET): omitting RT raised 10-yr local recurrence **9.5% vs 0.9%**, **OS identical** (80.8 vs 80.7%). **EUROPA** (≥70, luminal A): RT alone had better QoL than ET alone at interim — recurrence data pending
ALND still indicated	≥3 positive nodes, matted nodes, inflammatory/T4, **node-positive after neoadjuvant**

TREATMENT PATHWAY BY SUBTYPE

HR+ / HER2– early

Surgery first (neoadjuvant only if downstaging)
Oncotype on the specimen: N0 or 1–3 nodes
chemo by RS + menopausal status (TAILORx, RxPONDER)

Endocrine therapy for all, 5–10 yr
postmenopausal: AI · premenopausal: tamoxifen
high risk premenopausal: OFS + AI or tamoxifen

Escalate for high risk
≥4 nodes or 1–3 + high risk: abemaciclib ×2 yr
stage II–III incl. N0: ribociclib ×3 yr
gBRCA high risk: olaparib ×1 yr

HER2+ early

≤cT1 N0: surgery first
paclitaxel ×12 wk + trastuzumab ×1 yr (APT)

≥cT2 or N+: neoadjuvant
TCHP ×6 or **T-DXd ×4 → THP** (DB-11)

Adjuvant by response
pCR → complete 1 yr HP (pertuzumab if N+)
residual → T-DXd ×14 (DB-05) or T-DM1
HR+: neratinib ×1 yr option (ExteNET)

TNBC early

≥cT2 or N+: KEYNOTE-522
carbo/paclitaxel + pembro → AC + pembro
surgery → pembro to 1 yr, **regardless of pCR**

Residual disease
capecitabine (CREATE-X) · gBRCA: olaparib

Metastatic

HR+ / HER2–
1L: AI or fulvestrant + **CDK4/6i (ribociclib)**
2L: **by NGS** — ESR1, PIK3CA/AKT1/PTEN
then T-DXd (low/ultralow), SG, Dato-DXd

HER2+
1L: **T-DXd + pertuzumab** (DB-09) or THP
2L T-DXd · CNS: tucatinib-cape-trastuzumab

TNBC
CPS ≥10: pembro + chemo or **SG + pembro**
CPS <10: **SG or Dato-DXd** · gBRCA: PARPi
later: SG · T-DXd if HER2-low

RECEPTOR DEFINITIONS — ASCO/CAP

Result	Meaning
ER / PR ≥1%	HR-positive. **ER-low (1–10%)** behaves like TNBC — consider TNBC-type chemo/IO
HER2 IHC 0 — null	No membrane staining. No HER2-directed therapy
HER2-ultralow *IHC 0 with staining*	**Faint, incomplete membrane staining in ≤10% of cells**. T-DXd-eligible in HR+ mBC after ET (DESTINY-Breast06)
HER2-low	**IHC 1+** (faint, incomplete in >10%) **or IHC 2+ / ISH–**. Not HER2-positive — no trastuzumab benefit — but T-DXd works (DESTINY-Breast04/06)
HER2-positive	**IHC 3+** (complete, intense, >10%) **or IHC 2+ with ISH amplification**

Pathology reports must now distinguish 0-null from ultralow — ask for a re-read of an archival "IHC 0" before ruling out T-DXd. Scoring 0 vs 1+ is poorly reproducible; retest a metastatic biopsy.

HER2-LOW / ULTRALOW — T-DXD

Trial	Result
DESTINY-Breast04 *after 1–2 chemo lines*	HER2-low, HR+ or HR–: OS **23.4 vs 16.8** (HR+ cohort 23.9 vs 17.5; PFS 10.1 vs 5.4)
DESTINY-Breast06 *after ≥1 ET, chemo-naive*	HR+, low + ultralow: PFS **13.2 vs 8.1** (HR 0.62); OS immature
ILD	~11–12% in both, **with deaths**

GERMLINE TESTING

- **ASCO/SSO 2024: offer BRCA1/2 testing to every patient diagnosed at ≤65** (stage I–III or de novo IV). Over 65: if TNBC, male, PARP-inhibitor candidate, or suggestive personal/family history. NCCN uses ≤50 plus family/personal criteria
- **Testing is treatment-changing:** adjuvant olaparib (OlympiA), PARP inhibitor in mBC, and surgery choice (bilateral mastectomy, RRSO)
- **High penetrance: BRCA1** (TNBC-enriched) · **BRCA2** (HR+, male BC) · **PALB2** (manage like BRCA) · **TP53** (avoid RT if possible) · PTEN · CDH1 (lobular + diffuse gastric)
- **Moderate penetrance: CHEK2, ATM** — annual MRI screening; **risk-reducing mastectomy is not indicated on the gene alone**; RT is safe
- Offer a multigene panel — not BRCA alone — when history suggests other genes

Breast cancer

EARLY HR+ / HER2–

Question	Evidence
Chemo? N0	**TAILORx:** RS 11–25 → ET alone non-inferior (9-yr iDFS 83.3 vs 84.3%). **Age ≤50 with RS 16–25: chemo helps**. RS ≥26 → chemo
Chemo? 1–3 nodes	**RxPONDER** (RS ≤25): **postmenopausal — no benefit**; premenopausal — 5-yr iDFS 93.9 vs 89.0% (likely an OFS effect)
OFS	**SOFT/TEXT:** OFS + exemestane best DFS. Final 15-yr: OS gain confined to high risk — **age <35: 82.5 vs 77.9 vs 68.1%** (E+OFS / T+OFS / T)
Abemaciclib ×2 yr *monarchE*	≥4 nodes, or 1–3 + grade 3, ≥5 cm or Ki-67 ≥20%. 7-yr iDFS 77.4 vs 70.9%; **OS now significant** (HR 0.84; 86.8 vs 85.0%)
Ribociclib ×3 yr *NATALEE*	Stage IIB–III, or IIA N0 with grade 3 (or grade 2 + high genomic risk/Ki-67 ≥20%). 5-yr iDFS HR 0.72 (+4.5 points); **OS not yet significant**
Olaparib ×1 yr *OlympiA*	gBRCA, high-risk HER2–. 6-yr OS 87.5 vs 83.2% (HR 0.72)

EARLY HER2+

Setting	Evidence
Small, N0	**APT** (single-arm, ≤3 cm): TH → 10-yr iDFS 91.3%. **ATEMPT:** T-DM1 ×1 yr — 3-yr iDFS 97.8% but **no less toxicity** (46 vs 47%)
Neoadjuvant	NeoSphere: pertuzumab raised pCR 29 → 46%. **DESTINY-Breast11:** T-DXd ×4 → THP vs ddAC → THP, pCR **67.3 vs 56.3%**, grade ≥3 AE 38 vs 56%; **T-DXd alone was inferior** and closed
Residual disease	**KATHERINE:** T-DM1 ×14, iDFS HR 0.50, 7-yr OS 89.1 vs 84.4%. **DESTINY-Breast05** (high risk): **T-DXd vs T-DM1**, 3-yr iDFS **92.4 vs 83.7%** (HR 0.47); ILD 9.6%
Pertuzumab adjuvant	**APHINITY:** iDFS HR 0.81; benefit is **node-positive only** (8-yr 86.1 vs 81.2%); **OS not significant**
Extended	**ExteNET:** neratinib ×1 yr, 5-yr iDFS 90.2 vs 87.7%; gain in HR+; **no OS benefit**; 40% grade 3 diarrhea without prophylaxis

EARLY TNBC

Trial	Evidence
KEYNOTE-522	Stage II–III: pCR **64.8 vs 51.2%**; **5-yr OS 86.6 vs 81.7%**. Give pembrolizumab regardless of PD-L1 or pCR
CREATE-X	Residual HER2– disease: capecitabine; TNBC OS HR 0.52
OlympiA	gBRCA with residual TNBC or ≥pT2/pN1: olaparib ×1 yr

METASTATIC HR+ — 1L CDK4/6 INHIBITORS

Trial	PFS (mo)	OS (mo)
MONALEESA-2 ribo + letrozole	25.3 vs 16.0	**63.9 vs 51.4**, HR 0.76 — sig
MONALEESA-3 ribo + fulvestrant	20.5 vs 12.8	**53.7 vs 41.5**, HR 0.73 — sig
MONALEESA-7 ribo, premeno	23.8 vs 13.0	42-mo OS 70.2 vs 46.0%, HR 0.71 — sig
MONARCH 2 abema + fulvestrant	16.4 vs 9.3	**46.7 vs 37.3**, HR 0.76 — sig
MONARCH 3 abema + NSAI	28.2 vs 14.8	66.8 vs 53.7, HR 0.80 — **NS**
PALOMA-2 / -3 palbo + letrozole / fulv	24.8 vs 14.5 / 9.5 vs 4.6	**NS** in both

Ribociclib is the only CDK4/6i with OS benefit in every phase 3 — the preferred 1L agent. HR+/HER2+ mBC: PATINA added palbociclib to maintenance HP + ET after induction — PFS 44.3 vs 29.1 mo (HR 0.75), OS pending.

METASTATIC HR+ — SEQUENCING

- **Endocrine-sensitive** (de novo, or relapse >12 mo after adjuvant ET) → **AI + CDK4/6i**. **Endocrine-resistant** (on or ≤12 mo after adjuvant ET) → **fulvestrant + CDK4/6i**, or the **inavolisib triplet if PIK3CA-mutant**
- **Visceral crisis** → chemotherapy first; otherwise endocrine-based therapy precedes chemo. **Premenopausal/men: add GnRH agonist**
- **Re-test ctDNA at each progression** (ESR1 acquired on AI)

METASTATIC HER2+

Line	Evidence
1L	**DESTINY-Breast09:** T-DXd + pertuzumab vs THP, PFS **40.7 vs 26.9** (HR 0.56), OS immature, ILD 12%. **CLEOPATRA** (THP): OS **57.1 vs 40.8** — still reasonable
2L	**DESTINY-Breast03:** T-DXd vs T-DM1, PFS **28.8 vs 6.8**, OS HR 0.64
Brain mets	**HER2CLIMB:** tucatinib + trastuzumab + capecitabine, OS 21.9 vs 17.4, **CNS-PFS 9.9 vs 4.2**, active mets allowed · T-DXd active (DB-12)

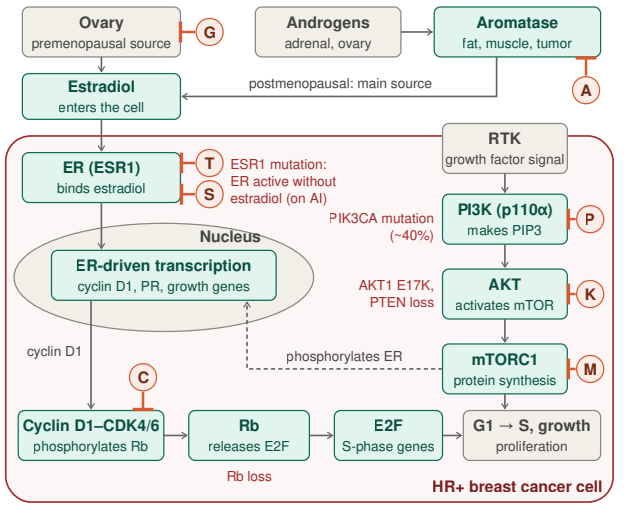
METASTATIC TNBC

Setting	Evidence
1L CPS ≥10 *22C3*	**Pembro + chemo** (KEYNOTE-355): OS **23.0 vs 16.1**, HR 0.73; **no OS gain at CPS 1–9** · **SG + pembro** (ASCENT-04): PFS 11.2 vs 7.8, OS immature
1L, not IO candidate	**Dato-DXd** (TROPION-Breast02): PFS 10.8 vs 5.6, **OS 23.7 vs 18.7** · **SG** (ASCENT-03): PFS 9.7 vs 6.9, OS immature
gBRCA	**Olaparib** (OlympiAD, PFS 7.0 vs 4.2) or **talazoparib** (EMBRACA, 8.6 vs 5.6) — **neither improved OS**
2L+	**Sacituzumab** (ASCENT): OS 12.1 vs 6.7 · T-DXd if HER2-low (DB-04)

METASTATIC HR+ — AFTER CDK4/6 BY BIOMARKER

Target	Agent → evidence
ESR1 mut	**Elacestrant** (EMERALD): ESR1m PFS 3.8 vs 1.9 (HR 0.55) · **imlunestrant** (EMBER-3): 5.5 vs 3.8 (HR 0.62); **+ abemaciclib 9.4 vs 5.5** (combo approved Sep 2026) · **vepdegestrant** (VERITAC-2, May 2026)
ESR1 on 1L ctDNA	**SERENA-6:** ESR1m emerges before progression → switch AI to **camizestrant**, keep CDK4/6i: PFS 16.0 vs 9.2 (HR 0.44). Approved Sep 2026; OS immature
PIK3CA mut	**Inavolisib + palbo + fulv** (INAVO120, endocrine-resistant): PFS 15.0 vs 7.3, **OS 34.0 vs 27.0 (HR 0.67)** · **alpelisib + fulv** (SOLAR-1): 11.0 vs 5.7, OS NS
PIK3CA / AKT1 / PTEN	**Capivasertib + fulv** (CAPItello-291): altered PFS 7.3 vs 3.1 (HR 0.50)
PIK3CA wild-type	**Gedatolisib + fulv ± palbo** (VIKTORIA-1, Jul 2026)
No target	Switch CDK4/6i + ET (MAINTAIN, postMONARCH) · **everolimus + exemestane** (BOLERO-2)
Endocrine-refractory	**T-DXd** if HER2-low/ultralow (DB-04/06) · **Sacituzumab** (TROPiCS-02): OS 14.4 vs 11.2 · **Dato-DXd** (TROPION-Breast01): PFS HR 0.63 · gBRCA: PARPi

PATHOPHYSIOLOGY



G goserelin · leuprolide: GnRH agonist (ovarian suppression) · T tamoxifen: SERM

A anastrozole · letrozole · exemestane: aromatase inhibitors

S fulvestrant · elacestrant · imlunestrant · camizestrant: ER degraders (SERDs)

C palbociclib · ribociclib · abemaciclib: CDK4/6 inhibitors · M everolimus: mTORC1 inhibitor

P alpelisib · inavolisib: PI3Kα inhibitors · K capivasertib: AKT inhibitor

ESR1 mutations arise on AIs and are rare at diagnosis: retest ctDNA at progression; oral SERDs still degrade mutant ER. Red text = resistance alterations.

Breast cancer

EARLY-STAGE DOSING

Regimen	Dosing
ddAC → T	Doxorubicin **60** + cyclophosphamide **600** q14d ×4 (G-CSF) → paclitaxel **175 q14d ×4** or **80 weekly ×12**
TC	Docetaxel **75** + cyclophosphamide **600** q21d ×4
KEYNOTE-522	Carboplatin **AUC 5** d1 + paclitaxel **80** d1, 8, 15 + **pembrolizumab 200 mg** q21d ×4 → AC (60/600) + pembro q21d ×4 → surgery → **pembro ×9** (1 yr total)
TCHP / **TCH**	Docetaxel **75** + carboplatin **AUC 6** + trastuzumab **8 → 6 mg/kg** ± pertuzumab **840 → 420 mg** q21d ×6; HER2 therapy to 1 yr
TH *APT*	Paclitaxel **80** + trastuzumab **4 → 2 mg/kg** weekly ×12, then trastuzumab **6 mg/kg q3wk** to 1 yr
AC → TH(P)	AC q14d ×4 → paclitaxel 80 weekly ×12 + trastuzumab ± pertuzumab. **Never HER2 agents concurrent with anthracycline**
T-DM1 / **T-DXd** *residual HER2+*	T-DM1 **3.6 mg/kg** q21d ×14 · T-DXd **5.4 mg/kg** q21d ×14 (DB-05)
Capecitabine	**1000–1250 mg/m² BID d1–14** q21d ×6–8
Abemaciclib *adjuvant*	**150 mg BID ×2 yr** + ET
Ribociclib *adjuvant*	**400 mg daily d1–21 q28d ×3 yr** + NSAI
Olaparib	**300 mg BID ×1 yr** (tablets)
Endocrine	Tamoxifen **20 mg** · anastrozole **1** · letrozole **2.5** · exemestane **25 mg** daily · goserelin **3.6 mg SQ q4wk** / leuprolide **3.75 mg q4wk**

Doses mg/m² unless stated. Neratinib: loperamide prophylaxis or dose escalation from the start.

METASTATIC DOSING

Regimen	Dosing
CDK4/6 inhibitors	**Ribociclib 600 mg** d1–21 q28d · **palbociclib 125 mg** d1–21 q28d · **abemaciclib 150 mg BID** continuous (200 BID alone) — with AI or fulvestrant
Fulvestrant	**500 mg IM** d1, 15, 29, then q28d. Premenopausal: add OFS to AI or fulvestrant
Oral SERDs	**Elacestrant 345 mg** daily · imlunestrant **400 mg** daily · camizestrant **75 mg** daily + CDK4/6i
Capivasertib	**400 mg BID, 4 days on** / **3 off** + fulvestrant
PI3Kα	**Inavolisib 9 mg** daily + palbociclib + fulvestrant · **alpelisib 300 mg** daily + fulvestrant
Everolimus + exemestane	**10 mg + 25 mg** daily; dexamethasone mouthwash for stomatitis
THP	Docetaxel **75** (or weekly paclitaxel 80) + trastuzumab 8 → 6 mg/kg + pertuzumab 840 → 420 mg q21d; stop taxane after ~6 cycles, continue HP
T-DXd	**5.4 mg/kg q21d** (HER2+, low, ultralow) ± pertuzumab 1L. Highly emetogenic — 3-drug prophylaxis
Tucatinib-cape-H	Tucatinib **300 mg BID** + capecitabine **1000 mg/m² BID d1–14** + trastuzumab 8 → 6 mg/kg q21d
Sacituzumab	**10 mg/kg d1, d8 q21d** (± pembro 200 q3wk)
Dato-DXd	**6 mg/kg q21d**
PARP inhibitors	Olaparib **300 mg BID** · talazoparib **1 mg** daily
Bone-directed	Zoledronic acid **4 mg IV q4wk** (q12wk after 1 yr) or denosumab **120 mg SQ q4wk**; dental review, Ca/vit D

SURGICAL, RADIATION AND SPECIAL SITUATIONS

- **Inflammatory (T4d): trimodality always** — neoadjuvant systemic therapy → modified radical mastectomy + ALND → chest wall + nodal RT, **even after pCR**; no BCS, delayed reconstruction
- **BCS relative contraindications:** Li-Fraumeni, active scleroderma/lupus, prior chest RT, large tumor-to-breast ratio. **Mastectomy** for gene carriers, prior mantle RT, or preference
- **Pregnancy:** surgery any trimester; anthracycline chemo from the 2nd trimester; **no endocrine, HER2, IO or RT until after delivery**
- **Male breast cancer:** mostly HR+, BRCA2-enriched; **tamoxifen preferred**; an AI needs a GnRH agonist
- **Lymphedema** ~5% after SLNB, highest after ALND + nodal RT · **BIA-ALCL** (textured implants): cytology, then remove capsule + implant

TOXICITY AND MONITORING

- **T-DXd ILD** (10–15% across trials, fatal cases): baseline + **serial chest CT** on therapy, ask about cough/dyspnea at every visit. **Grade 1: hold** until resolved (same dose if ≤28 days), consider steroids. **Grade ≥2: stop permanently** + prompt systemic steroids
- **HER2 agents — cardiac:** LVEF at baseline and **q3 months**. Hold trastuzumab for **≥16% absolute drop**, or below normal with ≥10% drop; recheck in 4 wk
- **Ribociclib: QTc** — ECG baseline (start only if QTcF <450) and ~day 14; electrolytes; **never with tamoxifen**; LFTs + CBC q2wk ×2 cycles
- **Abemaciclib: diarrhea** — loperamide at the first loose stool; CBC/LFTs q2wk ×2 mo; VTE, ILD. **Palbociclib:** neutropenia, CBC d1 + d15 of cycles 1–2
- **Alpelisib** / **capivasertib** / **inavolisib: hyperglycemia** — fasting glucose + HbA1c first, optimize diabetes; glucose weekly-to-biweekly in month 1; metformin first-line; rash (alpelisib: antihistamine prophylaxis)
- **Sacituzumab:** neutropenia (grade ≥3 ~50%), diarrhea; UGT1A1*28 homozygotes at higher risk. **Dato-DXd:** stomatitis and keratitis — steroid mouthwash + eye care
- **Tamoxifen:** VTE, endometrial cancer (annual gyn review); avoid paroxetine/fluoxetine (CYP2D6). **AI:** DEXA baseline + q1–2 yr
- **Surveillance:** H&P q6–12 mo ×5 yr, annual mammogram; no routine markers or imaging

KEY TRIALS

KEYNOTE-522	Pembro + neoadjuvant chemo, stage II–III TNBC — 5-yr OS 86.6 vs 81.7%
monarchE / **NATALEE**	Adjuvant abemaciclib (OS HR 0.84) / ribociclib (iDFS HR 0.72)
DESTINY-Breast04 / **06**	T-DXd, HER2-low after chemo / low + ultralow after ET
DESTINY-Breast03 / **09**	T-DXd 2L vs T-DM1 / 1L + pertuzumab vs THP
DESTINY-Breast05 / **11**	T-DXd for residual HER2+ disease / neoadjuvant T-DXd → THP
MONALEESA-2	Ribociclib + letrozole 1L — OS benefit
INAVO120	Inavolisib triplet, PIK3CA-mut — OS benefit
SERENA-6	ctDNA-guided switch to camizestrant before progression
ASCENT / **-04**	SG in refractory TNBC / + pembro 1L CPS ≥10